

Checkpoint 2: DataLoader and Diffusion Pipeline

Objective

Show that paired images enter the model correctly and that the diffusion training task is implemented.

Key Message

The data pipeline, noising process, and tests are ready before full model training.

Completed Work

- Implemented normalized paired image loading.
- Implemented timestep sampling and linear beta schedule.
- Added sanity tests for dataset pairing, diffusion, and training steps.

Verification and Controls

- Applied synchronized transforms to preserve pixel alignment between underwater inputs and clean references.
- Validated timestep ranges, noise tensor shapes, and reverse-sampling output bounds.
- Used smoke training and unit tests to detect data-model incompatibilities before full experiments.

Evidence

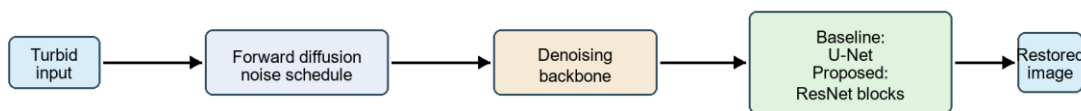
Component	Status
Paired DataLoader	Completed
Forward diffusion	Completed
Unit tests	Passing

Decision / Next Gate

Keep preprocessing, diffusion schedule, training budget, and callback logic fixed for all compared backbones.

Notes

- The same diffusion objective is used for both baseline and residual models.



Same paired dataset and evaluation protocol for both denoising backbones

Conditional diffusion restoration pipeline.